

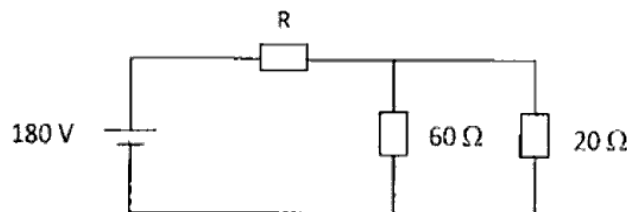
BE-104
B. E. (First/Second Semester), EXAMINATION, JUNE-2012
(Grading System)
(Common for all Branches)
BASIC ELECTRICALS & ELECTRONICS ENGINEERING
Time : Three Hours
Maximum Marks : 70
Minimum Pass Marks : 22 (D Grade)

Note : Attempt all the questions. All questions carry equal Marks.

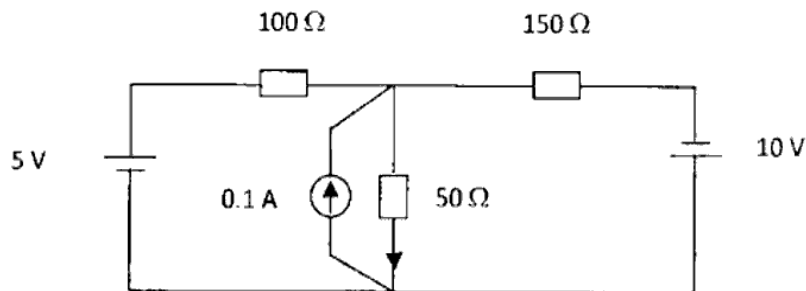
1. Answer any **Two** of the following :

(a) State the NORTON'S Theorem. In the circuit shown below determine

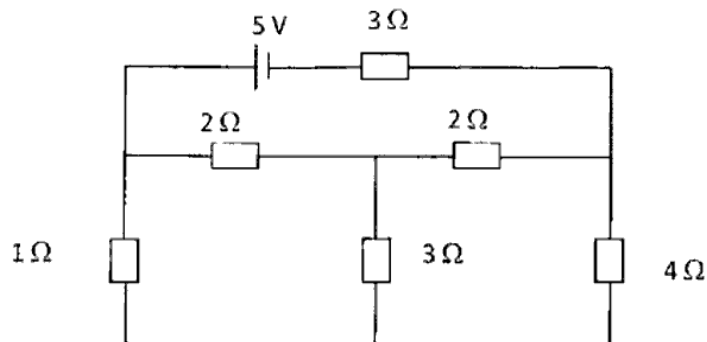
- (i) The value of R so that the load of $20\ \Omega$ draws maximum power.
- (ii) The value of maximum power drawn by the load.



(b) State superposition theorem. Apply the same for finding the current in $50\ \Omega$ resistor with the reference direction shown in circuit.



(c) Determine the current drawn from the 5 volt battery in the network shown.



(d) Explain the following terms : (i) RMS (ii) Average Value (iii) Active Power (iv) Reactive Power

P.T.O.

2. Answer any **two** of the following :

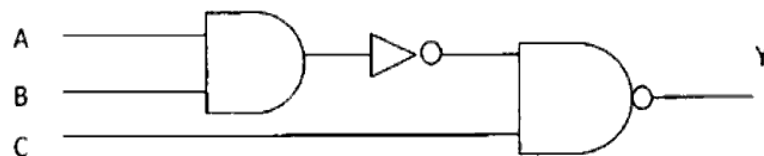
- (a) Explain the following w.r.t. transformer (i) Losses (ii) Voltage regulation
- (b) Draw the phasor diagram of a single phase transformer with an inductive load. Write down the procedure in steps for drawing the phasor diagram.
- (c) A 11 KV/400 V distribution transformer takes a no load primary current of 1 Amp at a power factor of 0.24 lagging. Find (i) The core loss current (ii) The magnetizing current (iii) The iron loss
- (d) Give reasons why :
 - (i) Rating of transformer is specified in KVA and not in KW.
 - (ii) Core losses are called iron losses. (iii) Cooling is required in transformer
 - (iv) Cores of transformer is laminated with laminated sheets

3. Answer any **two** of the following :

- (a) Draw torque-slip characteristics of a 3 phase induction motor. Explain the concept of slip.
- (b) Give reasons why :
 - (i) Starting current is high in dc motor (ii) Induced emf in a dc motor is called back emf
- (c) Specify the application of following motors in field (minimum two)
 - (i) Three phase induction motor (ii) Synchronous motor
 - (iii) DC motors (iv) Single phase induction motor
- (d) A 3 phase, 6 pole induction motor runs at 960 rpm on full load. It is supplied from a 4 pole alternator running at 1500 rpm. Calculate full load slip of the motor.

4. Answer any **two** of the following :

- (a) Obtain the following :
 - (i) Binary equivalent of $(123.72)_8$ (ii) Octal equivalent of $(10010110.1011)_2$
 - (iii) Hexadecimal equivalent of $(2391)_{10}$ (iv) Decimal equivalent to $(11011000)_2$
- (b) Draw the truth table of the following logic circuit



- (c) State and prove Demorgan's theorem using two variables.
- (d) Explain the operation of following flip flops.
 - (i) J-K Flip flop (ii) R-S Flip flop

5. Answer any **two** of the following :

- (a) Compare the CE, CB and CC configuration of BJT on the basis of -
 - (i) Input resistance (ii) Output resistance
 - (iii) Voltage gain (iv) Current gain
- (b) Specify the following terms :
 - (i) Forbidden energy gap (ii) Intrinsic semiconductor
 - (iii) Doping (iv) Charge carriers (v) Biasing
- (c) Explain the operation of BJT under following mode :
 - (i) Cut off mode (ii) Active mode (iii) Saturation mode
- (d) Distinguish the following :
 - (i) Semiconductor and Insulator (ii) P-Type and N-Type materials.